

The procedure we have outlined can be reversed if we assume that observables possess a Lie-Jordan structure. Given a real Lie-Jordan algebra $\mathfrak{A}$ with $c = \hbar^2$, requiring (14.5) produces an associative complex $*$ -algebra. We can perform this on $\mathfrak{A}_1 := \mathbb{C} \otimes \mathfrak{A}$: the involution $*$ is $(a \otimes A)^* := \bar{a} \otimes A$ for every $a \in \mathbb{C}$ and $A \in \mathfrak{A}$, and the associative product is given by extending (14.8) $\mathbb{C}$ -linearly.

It is important to notice that the Lie-Jordan algebra structure can be equipped with a natural norm. After completion, this will generate an associated C^* -algebra [AlSc03]. Coming back to physics, it is finally worth stressing that with this formulation both the non-commutativity and non-associativity underlying the algebra of observables appear to have a quantum nature, and seem to realise quantum *deformations* of classical structures, in the sense that commutativity and associativity are restored as soon as $\hbar \rightarrow 0$. However, a relic of quantum non-commutativity remains in the Poisson bracket even at classical level. As a matter of fact, the picture we have described can be developed further. The (algebraic) *deformation quantisation* procedure, introduced in §11.5.8, starts by assuming that (14.7) is just the first-order expansion in $\hbar$ of the quantum product. This approach has been exploited in algebraic Quantum Field Theory in a modern and powerful fashion to reformulate the perturbative interacting theory, including the renormalisation procedure [BrFr09].

14.1.3 The GNS reconstruction theorem

The set of algebraic states on $\mathfrak{A}_S$ is a convex subset in the dual $\mathfrak{A}'_S$ of $\mathfrak{A}_S$: if ω_1 and ω_2 are positive and normalised linear functionals, $\omega = \lambda \omega_1 + (1 - \lambda) \omega_2$ is clearly still positive and normalised, for any $\lambda \in [0, 1]$.

Hence, just as we saw for the standard formulation, we can define *pure algebraic states* as extreme elements of the convex body.

Definition 14.3. *An algebraic state $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ on the unital C^* -algebra $\mathfrak{A}$ is called a **pure algebraic state** if it is extreme in the set of algebraic states. An algebraic state that is not pure is called **mixed**.*

Later we will show that the space of states is non-empty and compact in the $*$ -weak topology. Consequently, pure states exist.

Surprisingly, most of the abstract apparatus given by a C^* -algebra and a set of states admits elementary Hilbert space representations once a reference algebraic state has been fixed. This is by virtue of a famous procedure Gelfand, Najmark and Segal came up with, that we present below [Haa96, Ara09, Str05a].

Theorem 14.4 (GNS theorem). *Let $\mathfrak{A}$ be a C^* -algebra with unit $\mathbb{I}$ and $\omega : \mathfrak{A} \rightarrow \mathbb{C}$ a positive linear functional with $\omega(\mathbb{I}) = 1$. Then*

(a) *there exist a triple $(H_\omega, \pi_\omega, \Psi_\omega)$, where H_ω is a Hilbert space, $\pi_\omega : \mathfrak{A} \rightarrow \mathfrak{B}(H_\omega)$ a $\mathfrak{A}$ -representation over H_ω and $\Psi_\omega \in H_\omega$, such that:*

- (i) Ψ_ω is cyclic for π_ω : the invariant subspace $\mathcal{D}_\omega := \pi_\omega(\mathfrak{A})\Psi_\omega$ is dense in H_ω ;*
- (ii) $(\Psi_\omega | \pi_\omega(a) \Psi_\omega) = \omega(a)$ for every $a \in \mathfrak{A}$.*

(b) *If (H, π, Ψ) satisfies (i) and (ii), there exists a unitary operator $U : H_\omega \rightarrow H$ such that $\Psi = U\Psi_\omega$ and $\pi(a) = U\pi_\omega(a)U^{-1}$ for any $a \in \mathfrak{A}$.*

Proof.

(a) For a start we will build the Hilbert space. We will refer to the elementary theory of Hilbert spaces of §3. Let us define the quadratic form $\langle x, y \rangle_\omega := \omega(x^*y)$, $x, y \in \mathfrak{A}$. This is a Hermitian semi-inner product by the requests made on ω . Observe in particular that $\overline{\langle x, y \rangle_\omega} = \langle y, x \rangle_\omega$ because $\overline{\omega(a)} = \omega(a^*)$ which is a consequence of linearity and $\overline{\omega(b)} = \omega(b)$ for $b = b^*$. The latter arises from $\mathbb{R}_+ \ni \omega((\mathbb{I}+b)(\mathbb{I}+b)) = \omega(\mathbb{I}) + 2\omega(b) + \omega(bb)$, since $\omega(\mathbb{I}) = 1 \in \mathbb{R}$ and $\omega(bb) \in \mathbb{R}_+$. So for the seminorm $p_\omega(x) := \sqrt{\langle x, x \rangle_\omega}$ the Schwarz inequality

$$\omega(x^*y) \leq \sqrt{\omega(x^*x)} \sqrt{\omega(y^*y)} \quad (14.9)$$

holds. Consider now the set $\mathcal{I}_\omega := \{x \in \mathfrak{A} \mid p_\omega(x) = 0\}$. Since p_ω is a seminorm, by (14.9) $\mathcal{I}_\omega$ is a subspace of $\mathfrak{A}$. As the symbol suggests, $\mathcal{I}_\omega$ is a left ideal in $\mathfrak{A}$, i.e. a subspace such that $yx \in \mathcal{I}_\omega$ for any $x \in \mathcal{I}_\omega$, $y \in \mathfrak{A}$. In fact by (14.9):

$$0 \leq p_\omega(yx)^4 = \omega(x^*y^*yx)^2 \leq \omega(y^*yx^*y^*y)\omega(x^*x) = 0.$$

Hence we may define the vector space $\mathcal{D}_\omega := \mathfrak{A}/\mathcal{I}_\omega$, quotient of $\mathfrak{A}$ by the ideal $\mathcal{I}_\omega$. The elements of $\mathcal{D}_\omega$ are thus cosets $[x]$ for the equivalence relation $\mathfrak{A}: x \sim y \Leftrightarrow x - y \in \mathcal{I}_\omega$. The vector space structure is naturally inherited by $\mathfrak{A}$, and makes $\alpha[x] + \beta[y] := [\alpha x + \beta y]$ meaningful¹, for any $\alpha, \beta \in \mathbb{C}$, $x, y \in \mathfrak{A}$. That $\mathcal{I}_\omega$ is a subspace guarantees the structure is well defined. Since $\mathcal{I}_\omega$ is also the left ideal of zeroes of the seminorm associated to the semi-inner product $\langle \cdot, \cdot \rangle_\omega$, as we just showed,

$$([x] | [y])_\omega := \langle x, y \rangle_\omega \quad \forall x, y \in \mathfrak{A} \quad (14.10)$$

is a well-defined Hermitian inner product on $\mathcal{D}_\omega$. Introduce the Hilbert completion H_ω of $\mathcal{D}_\omega$ for the inner product, which we continue to indicate with (14.10) on the entire H_ω . The representation π_ω is defined in the natural way on the dense subspace $\mathcal{D}_\omega = \mathfrak{A}/\mathcal{I}_\omega \subset H_\omega$ as:

$$(\pi_\omega(a))([b]) := [ab]. \quad (14.11)$$

Then $\mathcal{D}_\omega$ is by construction invariant under every $\pi_\omega(a)$. At last, let $\Psi_\omega := [\mathbb{I}]$. As $a \in \mathfrak{A}$ varies, the sets $\pi_\omega(a)\Psi_\omega = [a]$ fill the dense space $\mathcal{D}_\omega$. Therefore Ψ_ω is cyclic, as needed. It is easy to see that, by construction, $\mathfrak{A} \ni a \mapsto \pi_\omega(a)$ is linear on the dense domain $\mathcal{D}_\omega$ (hence it has an adjoint) and satisfies, for any $a, b, c \in \mathfrak{A}$, $\mu \in \mathbb{C}$:

- (i) $\pi_\omega(a)\pi_\omega(b) = \pi_\omega(ab)$,
- (ii) $\pi_\omega(a) + \pi_\omega(b) = \pi_\omega(a+b)$,
- (iii) $\mu\pi_\omega(a) = \pi_\omega(\mu a)$,
- (iv) $(\pi_\omega(b)\Psi_\omega | \pi_\omega(a)\pi_\omega(c)\Psi_\omega)_\omega = (\pi_\omega(a^*)\pi_\omega(b)\Psi_\omega | \pi_\omega(c)\Psi_\omega)_\omega$.

The last fact, equivalent to

$$\pi_\omega(a)^* \upharpoonright_{\mathcal{D}_\omega} = \pi_\omega(a^*),$$

follows by

¹ Notice that there is no identity like $[ab] = [a][b]$ in this construction. Only the linear structure, not the algebra, is constructed on the quotient.

$$\begin{aligned}
(\pi_\omega(b)\Psi_\omega | \pi_\omega(a)\pi_\omega(c)\Psi_\omega)_\omega &= ([b] | [ac])_\omega = \omega(b^*ac) = \omega((a^*b)^*c) \\
&= ([a^*b] | [c])_\omega = (\pi_\omega(a^*)\pi_\omega(b)\Psi_\omega | \pi_\omega(c)\Psi_\omega)_\omega.
\end{aligned}$$

By construction, for $a \in \mathfrak{A}$:

$$\omega(a) = \omega(\mathbb{I}^*a\mathbb{I}) = ([\mathbb{I}] | [a][\mathbb{I}])_\omega = (\Psi_\omega | \pi_\omega(a)\Psi_\omega)_\omega. \quad (14.12)$$

To finish (a) it is enough to prove that every operator $\pi_\omega(a) : \mathcal{D}_\omega \rightarrow \mathcal{H}_\omega$ is bounded, so it extends uniquely to a bounded operator on $\mathcal{H}_\omega$, because $\mathcal{D}_\omega \subset \mathcal{H}_\omega$ is dense. We will call the extended operators with the same names $\pi_\omega(a)$. Therefore properties (i)-(iv) will still be valid, by continuity. In particular, the operators being bounded, (iv) implies $\pi_\omega(a)^* = \pi_\omega(a^*)$. It is obvious that $\pi_\omega(\mathbb{I}) = I$ and so the map $\pi_\omega : \mathfrak{A} \rightarrow \mathfrak{B}(\mathcal{H}_\omega)$ will be a $*$ -representation.

To prove the boundedness of the $\pi_\omega(a)$, we begin by showing ω is continuous. We will only assume that the linear functional ω is positive, without using $\omega(\mathbb{I}) = 1$. If $h \in \mathfrak{A}$ is normal, since $|\sigma(h)| \leq \|h\|$ by the features of the spectral radius, theorem 8.36(c) gives $\sigma(h \pm \|h\|\mathbb{I}) \geq 0$. By theorem 8.25, $h \pm \|h\|\mathbb{I} = c^*c$, so the positivity and linearity of ω allows to say $\omega(h) \pm \|h\|\omega(\mathbb{I}) \geq 0$, meaning $|\omega(h)| \leq \omega(\mathbb{I})\|h\|$. In turn this implies ω is a bounded linear functional. In fact, if $y \in \mathfrak{A}$ is any element, y^*y is self-adjoint and so normal. Using the above result gives immediately $|\omega(y^*y)| \leq \omega(\mathbb{I})\|y^*y\|$. Finally, (14.9) with $x = \mathbb{I}$ says

$$|\omega(y)|^4 \leq \omega(\mathbb{I})^2 \|y^*y\|^2 = (\omega(\mathbb{I})\|y\|)^2,$$

hence $\|\omega\| \leq \omega(\mathbb{I})$. On the other hand, from $|\omega(\mathbb{I})| = \omega(\mathbb{I})$ and $\|\mathbb{I}\| = 1$ we have

$$\|\omega\| = \omega(\mathbb{I}).$$

In our case, as $\omega(\mathbb{I}) = 1$, we obtain $\|\omega\| = 1$.

If $\omega(x^*x) > 0$ we can repeat the argument for the linear functional

$$\mathfrak{A} \ni z \mapsto \rho(z) := \frac{\omega(x^*zx)}{\omega(x^*x)},$$

by construction linear, positive and such that $\rho(\mathbb{I}) = 1$; therefore $\|\rho\| = \rho(\mathbb{I}) = 1$. We conclude that the state ω satisfies

$$\omega(x^*y^*yx) \leq \|y^*y\|\omega(x^*x),$$

holding also when $\omega(x^*x) = 0$ because the Cauchy-Schwarz inequality forces $0 \leq \omega(x^*y^*yx) = \omega((x^*y^*y)x) \leq \sqrt{\omega((x^*y^*y)^*(x^*y^*y))} \sqrt{\omega(x^*x)}$. Consequently

$$\|(\pi_\omega(y))(x)\|_\omega = \|[yx]\|_\omega = \sqrt{\omega(x^*y^*yx)} \leq \sqrt{\|y^*y\|} \sqrt{\omega(x^*x)} \leq \|y\| \|x\|_\omega,$$

and so $\|\pi_\omega(y)\| \leq \|y\|$. This ends part (a).

(b) Just asking $U\pi_\omega(a)\Psi_\omega := \pi(a)\Psi$ for any $a \in \mathfrak{A}$ determines a densely-defined isometric operator, which we call U . This is well defined because $\pi_\omega(a)\Psi_\omega = \pi_\omega(a')\Psi_\omega$ implies $\pi(a)\Psi = \pi(a')\Psi$, as is evident from

$$\|\pi(a - a')\Psi\|^2 = \omega((a - a')^*(a - a')) = \|\pi_\omega(a - a')\Psi_\omega\|_\omega^2.$$

For the same reason U is isometric:

$$\|U\pi_\omega(a)\Psi_\omega\|^2 = \|\pi(a)\Psi\|^2 = \omega(a^*a) = \|\pi_\omega(a)\Psi_\omega\|_\omega^2.$$

Hence we can extend U to a continuous isometric operator on $\mathcal{H}$ with the same name. Similarly, let us construct an isometric operator $V : \mathcal{H} \rightarrow \mathcal{H}_\omega$ as the unique continuous extension of $V\pi(a)\Psi = \pi_\omega(a)\Psi_\omega$. By continuity, and using the density of $\pi(\mathfrak{A})\Psi$, we have $UV\Phi = \Phi$ for every $\Phi \in \mathcal{H}$. Therefore U is onto, beside isometric, and so unitary. That $\Psi = U\Psi_\omega$ and $\pi(a) = U\pi_\omega(a)U^{-1}$, $a \in \mathfrak{A}$, are obvious by construction. $\square$

The GNS theorem shows that given an algebraic state, the observables of $\mathfrak{A}$ are still represented by (bounded) self-adjoint operators on a Hilbert space $\mathcal{H}_\omega$, where the expectation value of ω takes the usual form $(\Psi_\omega|\pi(a)\Psi_\omega)$ with respect to a reference vector Ψ_ω . The latter vector allows to recover the whole Hilbert space by means of the representation π_ω itself, as we said in the GNS theorem (a), part (i). However the reader should notice that not all algebraic states on $\mathfrak{A}$ are represented by positive trace-class operators on $\mathcal{H}_\omega$, as we shall discuss shortly.

The representation π_ω need not be injective, i.e. faithful. From the proof and (14.11) with $b = \mathbb{I}$ in particular, we immediately have

Proposition 14.5. *The GNS representation $\pi_\omega : \mathfrak{A} \rightarrow \mathfrak{B}(\mathcal{H}_\omega)$ of the algebraic state ω on the unital C^* -algebra $\mathfrak{A}$ is faithful if the **Gelfand left ideal** of ω ,*

$$\mathcal{J}_\omega := \{a \in \mathfrak{A} \mid \omega(a^*a) = 0\},$$

is trivial, i.e., $\mathcal{J}_\omega = \{0\}$.

Algebraic states with trivial Gelfand left ideal are called **faithful**. Notice that the kernel $\text{Ker}(\pi_\omega)$ of the GNS representations π_ω of a C^* -algebra is a closed *two-sided* $*$ -ideal: it only depends on ω , not on the particular GNS triple, and generally it is smaller than $\mathcal{J}_\omega$. In fact, from (14.11) and the construction of the GNS representation, one easily sees that the elements of $\mathcal{J}_\omega$ are the elements $a \in \mathfrak{A}$ such that $\pi_\omega(a)\Psi_\omega = 0$. Conversely, $a \in \text{Ker}(\pi_\omega)$ satisfies $\pi_\omega(a)\pi_\omega(b)\Psi_\omega = 0$ for all $b \in \mathfrak{A}$, which is a much stronger requirement. Therefore, in principle ω can have faithful GNS representations even if $\mathcal{J}_\omega \neq \{0\}$ but just because $\text{Ker}(\pi_\omega)$ is trivial.

A technical result that was proved in passing, during the previous proof, and that is useful in itself, is the following.

Theorem 14.6 (Continuity of positive functionals). *If ω is a positive functional on the C^* -algebra $\mathfrak{A}$ with unit $\mathbb{I}$, then ω is continuous and $\|\omega\| = \omega(\mathbb{I})$.*

There is a useful technical corollary to the GNS theorem that deserves being stated and proved.

Corollary 14.7. *Let ω be an algebraic state on the unital C^* -algebra $\mathfrak{A}$ with associated GNS triple $(H_\omega, \pi_\omega, \Psi_\omega)$.*

(a) *If $\psi : \mathfrak{A} \rightarrow \mathbb{C}$ is linear, positive and $\psi \leq \omega$ ($\omega - \psi$ is positive), there exists a unique $T \in \mathfrak{B}(H_\omega)$ such that*

$$\psi(b^*a) = (\pi_\omega(b)\Psi_\omega | T\pi_\omega(a)\Psi_\omega)_\omega \quad \forall a, b \in \mathfrak{A}.$$

Moreover $0 \leq T \leq I$ and $T \in \pi_\omega(\mathfrak{A})'$ (T commutes with each $\pi_\omega(a)$, $a \in \mathfrak{A}$).

(b) *Conversely, if $0 \leq T \leq I$ and $T \in \pi_\omega(\mathfrak{A})'$, then $\psi(a) := (\Psi_\omega | T\pi_\omega(a)\Psi_\omega)_\omega$, for every $a \in \mathfrak{A}$, is a positive functional with $\psi \leq \omega$.*

Proof. (a) Take ψ as in the assumptions. Since

$$|\psi(b^*a)|^2 \leq \psi(b^*b)\psi(a^*a) \leq \omega(b^*b)\omega(a^*a) = \|[b]\|_\omega \|[a]\|_\omega,$$

setting $\psi'([b], [a]) := \psi(b^*a)$, Riesz's theorem warrants the existence of $T \in \mathfrak{B}(H_\omega)$ with $\psi'([b], [a]) = ([b] | T[a])_\omega$. In other terms $\psi(b^*a) = (\pi_\omega[b]\Psi_\omega | T\pi_\omega(a)\Psi_\omega)$. Furthermore, by construction:

$$([b] | (T\pi_\omega(a) - \pi_\omega(a)T)[c])_\omega = \psi(b^*ac) - \psi((a^*b)^*c) = \psi(b^*ac) - \psi(b^*ac) = 0.$$

(b) is immediate. $\square$

Remarks 14.8.

(1) The cyclic vector Ψ_ω is a unit vector, by (a) (ii) in the GNS theorem, since $a = \mathbb{I}$ and $\omega(\mathbb{I}) = 1$.

(2) Irrespective of the way one proves the GNS theorem, the $*$ -representation of unital C^* -algebras π_ω must be continuous, because of theorem 8.22, and must also satisfy $\|\pi_\omega(a)\| \leq \|a\|$ for any $a \in \mathfrak{A}$. In addition, the same theorem implies π_ω is isometric ($\|\pi_\omega(a)\| = \|a\|$ for any $a \in \mathfrak{A}$) precisely when it is faithful (one-to-one).

(3) If we deal with C^* -algebras *without unit*, algebraic states can be defined anyway. They are positive, bounded linear functionals, but $\omega(\mathbb{I}) = 1$ is replaced by $\|\omega\| = 1$. These two conditions are equivalent on unital C^* -algebras, by theorem 14.6. We saw in §7.6.5 that if we restrict to the C^* -algebra $\mathfrak{B}_\infty(H)$ of compact operators on a Hilbert space H (which has no unit, because in infinite dimensions the identity operator is never compact), algebraic states are exactly the positive operators of trace class with unit trace. $\blacksquare$

If ω is an algebraic state on $\mathfrak{A}$, every statistical operator on the Hilbert space of a GNS representation of ω , i.e. every positive, trace-class operator with unit trace $T \in \mathfrak{B}_1(H_\omega)$ (a *statistical operator*), determines an algebraic state

$$\mathfrak{A} \ni a \mapsto \text{tr}(T\pi_\omega(a)),$$

evidently. This is true, in particular, for $\Phi \in H_\omega$ with $\|\Phi\|_\omega = 1$, in which case the above definition reduces to

$$\mathfrak{A} \ni a \mapsto (\Phi | \pi_\omega(a)\Phi)_\omega.$$

To this end we have